STANDARD OPERATING PROCEDURE

Dual-Axis Silicone Mould Rotating Device

Balancing, operation, cleaning and maintenance of the dual-axis rotational casting machine for silicone moulds

Document number	SOP-BFL-ROT-01
Revision	1.0 (first issue). Supersedes the short operating procedure in Appendix E of the industrial training report
Effective date	18 October 2026
Next review	October 2027, after the first real mould has been balanced, or whenever the machine is modified
Equipment	Dual-axis silicone rotocaster: nested-box gimbal on a 750 × 450 mm plywood base, Arduino controller in a 3D-printed control box
Location	Biofluid Laboratory (Dr. Zahran Lab), Department of Biomedical Engineering and Health Sciences, Faculty of Electrical Engineering, UTM Johor Bahru
Prepared by	Mohamed Alamin Ahmed Yousif Ahmed, industrial trainee
Approved by	Dr. Ahmad Zahran Md Khudzari, Senior Lecturer and laboratory head
Related documents	Visual SOP poster (Appendix A); Technical Project Report TR-BFL-ROT-01; silicone TDS and SDS

	Name	Signature	Date
Prepared	Mohamed Alamin Ahmed Yousif Ahmed		
Reviewed			
Approved	Dr. Ahmad Zahran Md Khudzari		

WARNING The worm gearmotors stall silently and can burn out within seconds if a frame is out of balance or blocked. Balance every new mould before power is applied (Section 9.2), and switch to STOP at once if a frame stops while its motor is powered.

1 Purpose

This procedure describes how to prepare, balance, fill, run, shut down, clean and maintain the dual-axis silicone rotocaster in the Biofluid Laboratory. The machine turns a closed mould about two perpendicular axes so that liquid silicone coats every part of the inner wall while it gels, producing hollow parts such as organ and vessel models without a core. Following this procedure protects the user, the mould and the machine, and gives repeatable castings.

2 Scope

- Applies to every user of the rotocaster, including staff, postgraduate and undergraduate students and industrial trainees.
- Covers pre-use inspection, mould mounting, static balancing, silicone filling, start-up, running, shutdown, demoulding, cleaning, maintenance, commissioning after repair, fault finding and emergencies.
- Applies to moulds that follow the mould interface in Section 6.3. Moulds that do not follow it must be approved by the laboratory head before use.
- Does not cover firmware changes or mechanical modifications. These must be approved and recorded in the revision history.

3 Responsibilities

Table 1 Responsibilities

Role	Responsibilities
Laboratory head (Dr. Ahmad Zahran Md Khudzari)	Approves this SOP. Authorises trained users. Approves new mould designs and any mould above 5 kg. Releases the machine after a fault.
Trained operator	Follows this SOP. Weighs and balances each mould. Completes the checklists and run log. Stops the machine and reports faults and near misses.
Technician or maintainer	Carries out maintenance and the commissioning procedure after repairs. Keeps spare motors, bearings, fuses and printed parts.
New or untrained user	Operates only under the direct supervision of a trained operator until signed off in Section 14.

4 References

- Project brief and product design specification (PDS items a to h): Dual-Axis Silicone Mould Rotating Device, issued by Dr. Ahmad Zahran Md Khudzari.
- Technical Project Report TR-BFL-ROT-01: Dual-Axis Silicone Mould Rotating Device.
- Silicone technical data sheet (TDS) and safety data sheet (SDS).
- STMicroelectronics L298 dual full-bridge driver datasheet; Arduino Uno R3 documentation.
- UTM laboratory safety rules and Faculty of Electrical Engineering laboratory regulations.

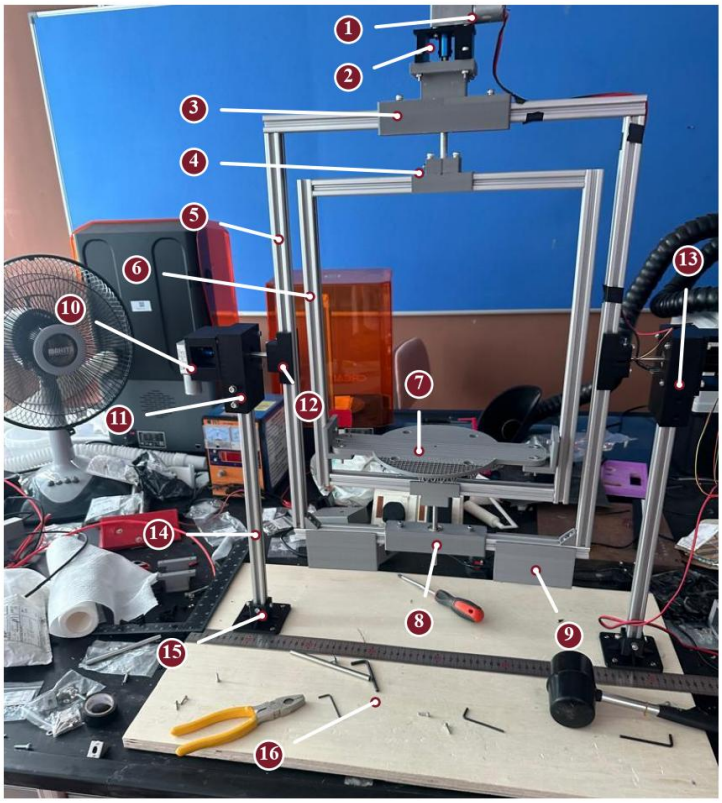
5 Definitions and Abbreviations

Table 2 Definitions and abbreviations

Term	Meaning
Outer axis	The horizontal axis. The outer frame turns on two trunnion shafts in the tower posts, driven by Motor 1.
Inner axis	The axis of the inner frame, perpendicular to the outer axis, driven by Motor 2 on the outer frame.
Static balance	The state in which a frame stays where it is put when released, because its centre of gravity lies on its axis.
Counterweight (CW)	One of two printed counterweights on the bottom rail at x = ±120 mm. Each holds M8 nuts clamped by a screw and balances Motor 2.
Slip ring	Six-wire 22 mm capsule slip ring on the right tower that feeds Motor 2 without winding the cable.
PCD	Pitch circle diameter of a bolt pattern.
PWM / duty	Pulse-width modulation. The duty cycle (0 to 100 %) sets the motor voltage and so its speed.
Gel time / cure time	Time after mixing when the silicone stops flowing / reaches handling strength.
OLED	Organic LED display, 1.3 inch, 128 × 64 pixels.

6 Equipment Description

The machine is a nested-box gimbal made of 2020 aluminium extrusion. The outer frame turns about a horizontal axis on two trunnion shafts carried by vertical tower posts. The inner frame turns inside it about the perpendicular axis, and the mould is bolted to a plate on the inner frame so that it sits where the two axes cross. Each axis has its own 12 V worm gearmotor. The gearboxes are self-locking, so the frames hold their position when the power is off. Figure 1 shows the machine as built.



Key

1 Motor 2, inner axis (JGY-370)

2 Motor 2 standoff and coupler (part 14)

3 Top bridge with Motor 2 pillar (part 03)

4 Inner-axis shaft gripper (part 02)

5 Outer frame, 2020 extrusion

6 Inner frame, 2020 extrusion

7 Mould plate (part 08) on plate shelves (part 06)

8 Bottom bridge (part 04)

9 Counterweights with M8 ballast (part 07, × 2)

10 Motor 1, outer axis, on its spacer (part 05)

11 Left tower bearing holder (part 01)

12 Outer-axis shaft gripper and trunnion (part 02)

13 Right tower with slip ring and Motor 2 wiring

14 Upright post, 2020 extrusion

15 Post socket (part 15)

16 Plywood base, 750 × 450 × 18 mm

Figure 1 Main parts of the dual-axis silicone rotocaster.

6.1 Operator controls

All controls are on the lid of the 3D-printed control box, which sits on the plywood base (Figure 2).

Table 3 Operator controls and indicators

Control or indicator	Function
RUN/STOP rocker switch (RUN)	RUN enables both motors. STOP stops both at once; the worm gears hold the frames.
OUTER potentiometer	Sets the speed of Motor 1 (outer axis). Fully anticlockwise = off.
INNER potentiometer	Sets the speed of Motor 2 (inner axis). Fully anticlockwise = off.
OLED display	Top line: RUN or STOP. Lower lines: duty cycle in % and the estimated speed in rpm for each axis.
12 V DC adapter	Supplies the controller through a 2 A fuse. Unplugging it isolates the machine.
USB port (left wall)	For uploading the control sketch only. Not used in normal operation.



(a) Body: board standoffs, USB, power and cable holes (b) Lid, outer face: OLED window, pots and switch

Figure 2 Control box: (a) body with board standoffs, USB, power and cable openings; (b) lid with OLED window, potentiometers and RUN switch.

NOTE There is no speed sensor. The displayed rpm is estimated from the duty cycle and the 18 rpm motor rating, so it reads high under load. Use a tachometer or count revolutions with a stopwatch when an exact speed is needed.

6.2 Main specifications and limits

Table 4 Specifications and operating limits

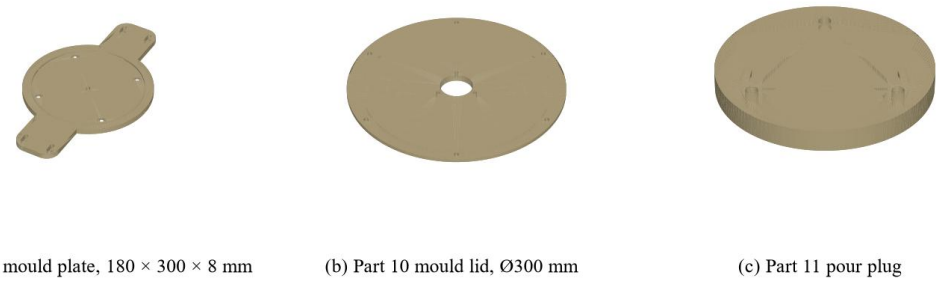
Item	Specification
Rotation	Two perpendicular axes, 360° continuous, independent speed control
Mould envelope	Ø300 × 300 mm (up to 380 mm high); clear opening of the inner frame Ø320 mm
Mould mass	**5 kg maximum** with the fitted JGY-370 motors. Up to 10 kg only after a 70 kg·cm motor is fitted and the laboratory head approves
Drives	2 × JGY-370 4632 worm gearmotors, 12 V, 18 rpm no-load (about 13 rpm loaded), 13 kg·cm (1.27 N·m), self-locking
Recommended casting speed	About 10–15 rpm per axis. Never above about 77 rpm, where silicone is pinned to the wall (centrifugal casting)
Soft start	15 s ramp from zero to the set speed; stopping is immediate
Controller	Arduino Uno, L298N dual H-bridge, two potentiometers, ON-OFF-ON rocker switch, 1.3-inch I2C OLED
Power	12 V DC wall adapter, 2 A fuse in the positive line; Motor 2 fed through a 6-wire slip ring (2 A per circuit)
Bearings and shafts	608ZZ bearings; Ø8 mm hardened steel shafts with D-flats; hollow Ø8 mm right trunnion carries the slip-ring wires
Footprint	Plywood base 750 × 450 × 18 mm; outer axis 375.8 mm above the base
Rotating mass without mould	About 5.35 kg

6.3 Mould interface

Moulds must follow this interface so that they fit the machine and can be balanced (Figure 3).

- **Mould foot:** bolted to the mould plate on a Ø140 mm PCD with 4 × M6 screws and washers. The plate slots allow ±6 mm of movement for balancing.
- **Mould rim:** bolted to the Ø300 mm lid on a Ø280 mm PCD with 6 × M5 screws, tightened in a star pattern.
- **Pour port:** Ø45 mm, closed by the pour plug with an O-ring and 3 × M4 screws; the plug cannot fall out when the mould is upside down.

- **Size:** the mould wall stays within about Ø125 mm at the base so the foot screws clear it, and the rim is flush at Ø300 mm so it never overhangs the Ø320 mm opening.



(a) Part 08 mould plate, 180 × 300 × 8 mm (b) Part 10 mould lid, Ø300 mm (c) Part 11 pour plug

Figure 3 Mould interface parts: (a) mould plate, (b) mould lid and (c) pour plug.

7 Health and Safety

WARNING The frames swing through the full space around the machine. Keep at least 0.5 m clear on all sides, keep hands, hair and cables out of the frame while it is powered, and never try to stop a frame by hand.

Table 5 Hazards and controls

Hazard	Possible harm	Control measures
Rotating outer and inner frames	Impact, entanglement of hair, sleeves or cables	0.5 m keep-out zone. Tie back hair, roll up sleeves. Switch to STOP before reaching in.
Pinch points at the shaft grippers, plate shelves and towers	Crushed or pinched fingers	Turn frames by hand only with the switch on STOP and the adapter unplugged.
Unbalanced frame	Sudden swing when released; motor stall and burnout	Balance before power (9.2). Lock the axis not being tested. Watch the first revolutions.
Mould or lid coming loose	Falling mould, silicone spill	Use the specified screws and washers; star-pattern tightening; check before each run.
Heavy mould	Motor overload, strain injury when lifting	Weigh every mould; 5 kg limit; lift with two hands close to the body.
Liquid silicone and catalyst	Skin and eye irritation	Nitrile gloves, safety glasses, good ventilation; follow the SDS.
Electrical supply and motor stall	Overheating, burning smell, fire	2 A fuse fitted; never bypass. Switch to STOP and unplug on any smell, smoke or stall.

7.1 Personal protective equipment

- **Nitrile gloves** for mixing, pouring, demoulding and cleaning. Keep gloved hands out of the frame while it is powered.
- **Safety glasses** for mixing, pouring and demoulding.
- **Laboratory coat** with fastened sleeves, **closed shoes**, and long hair tied back.

8 Materials and Tools

Table 6 Materials and tools

Item	Use
Silicone (two-part) with its TDS and SDS	Casting material
Digital scale (0.1 g) and a scale for the mould (up to 10 kg)	Weighing silicone and the filled mould

Item	Use
Mixing cups, stirrers, vacuum chamber (if available)	Mixing and de-gassing
Mould with foot, lid and pour plug to Section 6.3	Shapes the casting
M6 and M5 screws with washers; 3 × M4 for the plug; O-ring	Mounting the mould and closing it
Hex keys 2.5, 3, 4 and 5 mm; small screwdriver	Mounting, counterweights, grub screws
Spare M8 nuts (ballast)	Adjusting the counterweights
Strap or pin to lock a frame	Locking the axis not being balanced
Stopwatch or tachometer	Checking the real speed
Current-limited bench supply (12 V)	First run after any repair or a new mould (Appendix C)
Wipes and isopropyl alcohol	Cleaning

9 Procedure

9.1 Pre-use inspection (switch on STOP, adapter unplugged)

Table 7 Pre-use inspection

Step	Action	Acceptance check
1.1	Place the machine on a level, stable bench with 0.5 m clear on all sides.	No rocking; frame path clear
1.2	Check that both tower posts are locked in their base sockets with the M5 side screws.	Posts do not move
1.3	Check the frame corner brackets, shaft grippers, plate shelves and Motor 1 spacer for loose screws or cracks.	All tight, no cracks
1.4	Check that both counterweight screws are tight, the M8 nuts are clamped and the covers are fitted.	Nuts do not rattle; ballast cannot fall out
1.5	Check the slip-ring bracket, the coupler pinch screws and the wires to Motor 2.	Wires secure, not rubbing on the frame
1.6	Inspect the adapter, cable, fuse holder and the motor cables leaving the control box.	No damage; fuse fitted
1.7	Set the switch to STOP and both potentiometers fully anticlockwise.	STOP, both at zero

9.2 Mounting and balancing the mould

The worm gears are self-locking, so the frames cannot be turned by hand while the motors are coupled. Balance with the drive of the axis being checked disengaged (loosen the coupler grub screws), or by running that axis very slowly and watching whether it labours at one point of the turn. Retighten the grub screws on the D-flat afterwards.

Table 8 Mounting and static balancing

Step	Action	Acceptance check
2.1	Weigh the mould, including the silicone that will be poured.	Mass ≤ 5 kg; recorded in the run log
2.2	Bolt the mould foot to the mould plate with 4 × M6 screws and washers on the Ø140 mm PCD. Nip the plate's M4 slot screws only.	Mould seated, plate can still slide
2.3	Lock the inner frame. Disengage the outer drive, turn the outer frame to 90° and release it.	Frame stays where it is put
2.4	If it swings back, the heavy side is the one that ends at the bottom. Adjust the M8 nuts in the counterweights as in 9.2.1, then repeat 2.3.	Frame holds at 90° and at 0°
2.5	Lock the outer frame. Disengage the inner drive, turn the inner frame to 90° and release it. Correct by sliding the mould plate in its ±6 mm slots.	Inner frame holds its position
2.6	Tighten the plate M4 screws and the counterweight covers. Re-engage both drives: grub screws on the D-flats.	All screws tight; drives engaged

9.2.1 Adjusting the counterweights (outer axis)

The two counterweights on the bottom rail of the outer frame balance Motor 2 at the top of the frame. Their mass is set by the number of M8 nuts clamped in each counterweight by its screw. Use the result of the 90° hold test to decide which way to go:

- **Motor 2 (top) swings down:** the frame is top-heavy. **Add** M8 nuts to the counterweights.
- **The counterweights (bottom) swing down, or the frame rests with Motor 2 at the top:** the frame is counterweight-heavy. **Remove** M8 nuts. This is the fault that burnt out Motor 1 in the first powered test.

Table 9 Adjusting the counterweight ballast

Step	Action	Acceptance check
CW1	Set the switch to STOP and unplug the adapter. Hold the outer frame, or lock it, so it cannot swing while the ballast is changed.	Frame cannot move
CW2	Turn the outer frame so the counterweights are easy to reach. Loosen the counterweight screw with a hex key until the nuts are free. Do not remove the counterweight from the rail.	Nuts free, counterweight still on the rail
CW3	Add or remove M8 nuts (about 5 g each), one or two at a time. Keep the same number of nuts in both counterweights.	Equal nuts left and right
CW4	Tighten the counterweight screw firmly so the nuts are clamped. Refit the cover if it was removed.	Nuts do not move or rattle when the counterweight is shaken
CW5	Repeat the 90° hold test (step 2.3). Continue CW1 to CW4 until the frame holds its position at 90° and at 0°.	Frame holds its position
CW6	Record the final number of nuts per counterweight and the mould ID in Form A.	Value recorded for the next run with this mould

WARNING The counterweights turn upside down on every revolution. Never run the machine with a loose counterweight screw or loose nuts: they can shift, unbalance the frame and stall the motor, or fall out.

CAUTION A centred mould adds mass but almost no unbalance; an off-centre mould quickly uses up the motor margin. At 10 kg, a 10 mm offset alone needs about 1 N·m, close to the 1.27 N·m the fitted motors can give.

9.3 Filling the mould

Table 10 Filling

Step	Action	Acceptance check
3.1	Read the TDS and record the mix ratio, pot life, gel time and cure time.	Values in the run log
3.2	Weigh and mix the silicone; de-gas it if a vacuum chamber is available.	Correct ratio, no streaks
3.3	Fit the lid on the mould rim with 6 × M5 screws and washers on the Ø280 mm PCD, tightening in a star pattern.	Lid seated evenly
3.4	Pour the measured silicone through the Ø45 mm port. Fit the pour plug with its O-ring and 3 × M4 screws.	No leaks when the mould is tilted
3.5	Wipe any silicone off the outside of the mould and the plate.	Mould clean

9.4 Start-up and running

Table 11 Start-up and running

Step	Action	Acceptance check
4.1	Confirm the switch is on STOP and both potentiometers are at zero. Plug in the 12 V adapter.	OLED shows ROTOCASTER CONTROLLER, then STOP
4.2	Set the switch to RUN.	OLED shows RUN; motors still at 0 %

Step	Action	Acceptance check
4.3	Turn the INNER potentiometer up slowly to the set speed.	Inner frame starts smoothly and ramps up over 15 s
4.4	Turn the OUTER potentiometer up slowly to the set speed.	Outer frame turns smoothly, no labouring at the top
4.5	Watch the first five revolutions of each frame from outside the keep-out zone.	No stall, knock, rub, cable pull or leak
4.6	Keep the machine running for at least the gel time of the silicone. Check it regularly.	Continuous, steady rotation

Speed setting. Rotational casting works by letting the silicone tumble over the mould wall, so the speed must stay well below the point at which centrifugal force holds it against the wall. For the Ø300 mm mould envelope (r = 0.15 m) that point is

$$n_c = (60 / 2\pi)\sqrt{(g / r)} = 9.549 \times \sqrt{(9.81 / 0.15)} \approx 77\ rpm \quad (1)$$

Use about 10–15 rpm on each axis. Setting the two axes to different speeds avoids a repeating path and gives a more even wall. Record the duty cycles used, so a good result can be repeated.

WARNING If a frame stops, slows sharply or hums while its motor is powered, set the switch to STOP at once. A stalled worm gearmotor overheats and burns out within seconds. Re-balance before restarting.

9.5 Shutdown and demoulding

Table 12 Shutdown and demoulding

Step	Action	Acceptance check
5.1	Set the switch to STOP and turn both potentiometers to zero.	Both frames stop and hold
5.2	Unplug the 12 V adapter.	OLED off
5.3	Leave the mould on the machine or on a level surface until the cure time has passed.	Silicone tack-free
5.4	Remove the lid screws in a star pattern, take the lid off and remove the casting. Unbolt the mould foot.	Casting complete; mould removed
5.5	Inspect the casting for thin areas, pools and bubbles, and record the result and speeds used.	Run log completed

9.6 Cleaning

- Clean only with the switch on STOP and the adapter unplugged.
- Wipe uncured silicone off the frames, plate and lid with dry wipes, then a little isopropyl alcohol. Peel off cured silicone.
- Keep liquids away from the control box, motors and slip ring. Do not spray the machine.
- Return both frames to their rest position, check the screws and bearings, and store the adapter cable tidily.

10 Emergency Procedures

Table 13 Emergency actions

Situation	Immediate action
Person struck or caught by a frame	Set the switch to STOP and unplug the adapter. Do not force the frame back by hand; loosen the coupler to free it. Give first aid and report.
Frame stalls or a motor hums or smells hot	STOP at once and unplug. Let the motor cool. Re-balance and test the motor on a current-limited supply before reuse.
Mould or lid comes loose; silicone spills	STOP and unplug. Contain the spill away from the control box and slip ring. Clean as in 9.6.
Smoke, sparking or fire	Unplug at the wall. Do not use water. Use a CO_2 extinguisher and raise the alarm.

Situation	Immediate action
Silicone in the eyes or on the skin	Eyes: rinse with water for 15 minutes and seek medical advice. Skin: wipe off and wash with soap and water. Follow the SDS.

WARNING After any emergency, tag the machine "OUT OF SERVICE" and report to the laboratory head. Run the commissioning procedure in Appendix C before it is used again.

11 Routine Maintenance

Table 14 Maintenance schedule

Interval	Task	By
Before every run	Pre-use inspection (9.1) and balance (9.2)	Operator
After every run	Clean the frames, plate and lid (9.6)	Operator
Monthly	Check all T-nut screws, corner brackets, post socket screws, shaft gripper screws and both counterweight screws; retighten as needed	Technician
Monthly	Check the coupler grub screws on both D-flats and the slip-ring coupler pinch screws	Technician
Monthly	Turn each frame by hand with the drive disengaged: listen for rough or noisy 608ZZ bearings; replace if rough	Technician
Every 3 months	Inspect printed parts for cracks, creep or wear around bearing seats and D-flats; reprint from the CAD files if damaged	Technician
Every 3 months	Check the slip-ring wiring and the Motor 2 cable for chafing; check motor current on the bench supply (about 0.35 A running)	Technician
Yearly	Review this SOP, the run log and the fault log	Laboratory head

12 Troubleshooting

Table 15 Troubleshooting guide

Problem	Likely cause	Action
OLED blank	Adapter unplugged, fuse blown, loose I2C wire	Check the adapter and fuse. If the fuse has blown, find the cause before replacing it with another 2 A fuse.
OLED picture shifted or noisy strip at one edge	Display controller is SSD1306, not SH1106	Technician: select the SSD1306 line in the sketch and re-upload.
OLED shows RUN but a motor does not turn	Potentiometer at zero or in the dead band; wire off the L298N; motor faulty	Raise the potentiometer. Check the motor terminals. Test the motor on a current-limited supply.
Motor 2 runs intermittently	Slip-ring contact or wire fault	Check the slip-ring wires and coupler. Replace the slip ring if contact is intermittent.
Frame labours or stops at the same point of each turn	Unbalance; counterweight nuts loose or wrong number	STOP. Repeat the balancing procedure (9.2) and adjust the counterweight nuts (9.2.1).
Frame turns but the mould wobbles	Plate or foot screws loose	STOP, unplug, tighten the M6 foot and M4 plate screws.
Knocking or grinding	Loose gripper or coupler; worn bearing	STOP, unplug, check grub screws and grippers, check bearings.
Displayed speed higher than measured	Displayed rpm is an estimate	Normal. Use a tachometer or stopwatch for the real speed.
Uneven wall: thick bands, thin poles	Speed too high or both axes at the same speed	Lower the speeds and set the axes to different speeds.
Bubbles or voids	Silicone not de-gassed; pour plug leaking	De-gas; check the O-ring and plug screws.

13 Records

Keep these forms in the machine folder. Completed forms are kept for at least two years.

Form A: Pre-use and balance checklist

Date	Operator	Posts & screws	CW nuts (each)	Mould mass (kg)	Outer holds 90°	Inner holds 90°	Sign

Form B: Run log

Date	Operator	Mould ID	Silicone & batch	Mass (g)	Inner / outer duty %	Run time	Result / remarks

14 Training and Competency

1. Read this SOP, the silicone SDS and the visual SOP poster.
2. Watch a trained operator balance a mould and complete a full run.
3. Balance a mould and complete one full run under supervision, including Forms A and B.
4. Be signed off by the laboratory head or a person nominated by them.

Name	Matric / staff no.	Trained by	Date	Signature

15 Revision History

Rev.	Date	Description	By
1.0	18 October 2026	First issue; expands the operating procedure in Appendix E of the industrial training report	M. Alamin A. Y. Ahmed

Appendix A Visual SOP

Print this page in colour on A4 and display it next to the machine. It summarises Sections 7 to 10.



Figure 4 Visual SOP poster for the dual-axis silicone mould rotating device.

Appendix B Load Chart

The table gives the torque the outer axis needs with a residual mould offset of 10 mm, compared with the 1.27 N·m rating of the fitted JGY-370 motor. The machine is limited by unbalance rather than speed, so careful balancing matters more than the speed setting.

Table 16 Torque demand on the outer axis and allowed mould mass

Mould mass	10 rpm	20 rpm	30 rpm	Safety factor with fitted motor	Allowed?
2 kg	0.22 N·m	0.24 N·m	0.26 N·m	5.7	Yes
5 kg	0.51 N·m	0.54 N·m	0.56 N·m	2.4	Yes (limit)
10 kg	1.01 N·m	1.04 N·m	1.07 N·m	1.2	No, unless a 70 kg·cm motor is fitted

Appendix C Commissioning After Repair or Motor Replacement

Use this sequence after any repair, a motor or wiring change, or a new firmware upload, so that a wiring fault cannot damage a motor or the frame.

Step	Action
C1	Upload the sketch over USB with the 12 V adapter and both motors unplugged.
C2	With a multimeter, check the 12 V line for shorts to ground, and that the ground is common to the Arduino, L298N, potentiometers and OLED.
C3	Apply 12 V with the motors disconnected. Check that the OLED shows RUN/STOP and follows both potentiometers.
C4	Connect Motor 1 off the frame and run it from a bench supply at 12 V with the current limit set to about 0.5 A. Repeat for Motor 2.
C5	Fit the motors, balance the frames (9.2) and run one axis at a time with the other locked, with a hand on the RUN switch.
C6	Record the work in the maintenance log and the fault log, then release the machine to the laboratory head.

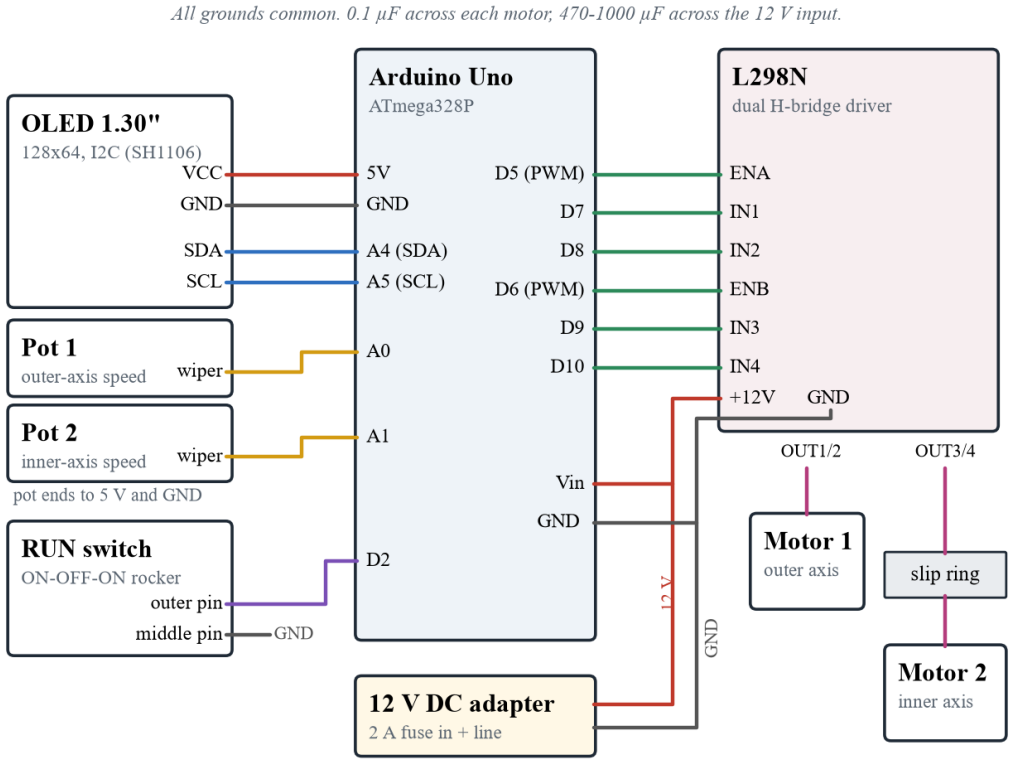


Figure 5 Wiring diagram of the controller, for reference during commissioning.